

Unpacking Spatial Dependence: A Variance-Stabilized Approach to Spatial Autoregressive Simulations

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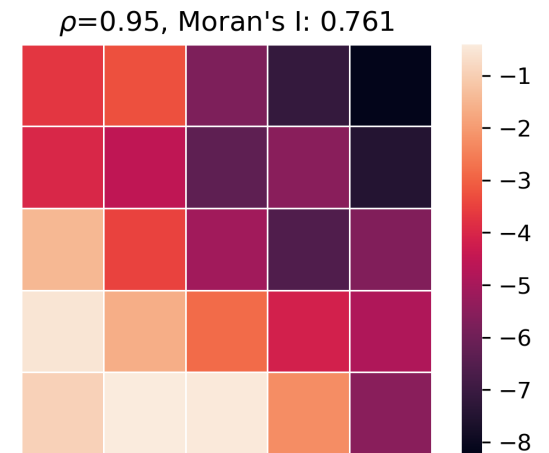
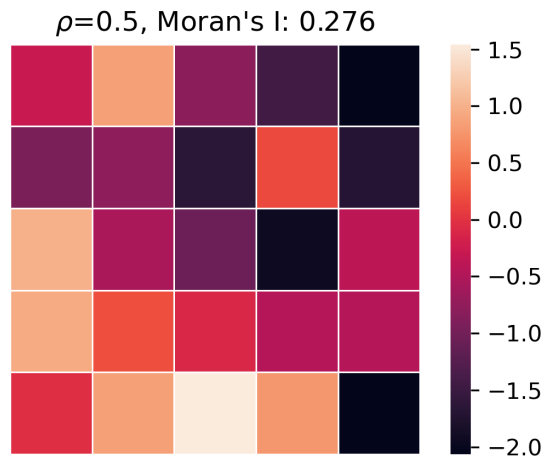
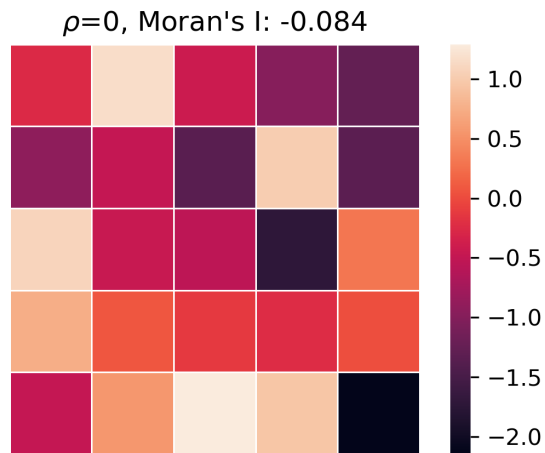
Simulating spatial patterns are key to testing models/stat learning algorithms, diagnostics, and estimators.

SAR is an exceptionally common DGP

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

- changing ρ changes both spatial autocorrelation and variance.
- misinterpretation: results may be driven by heteroskedasticity, not autocorrelation.

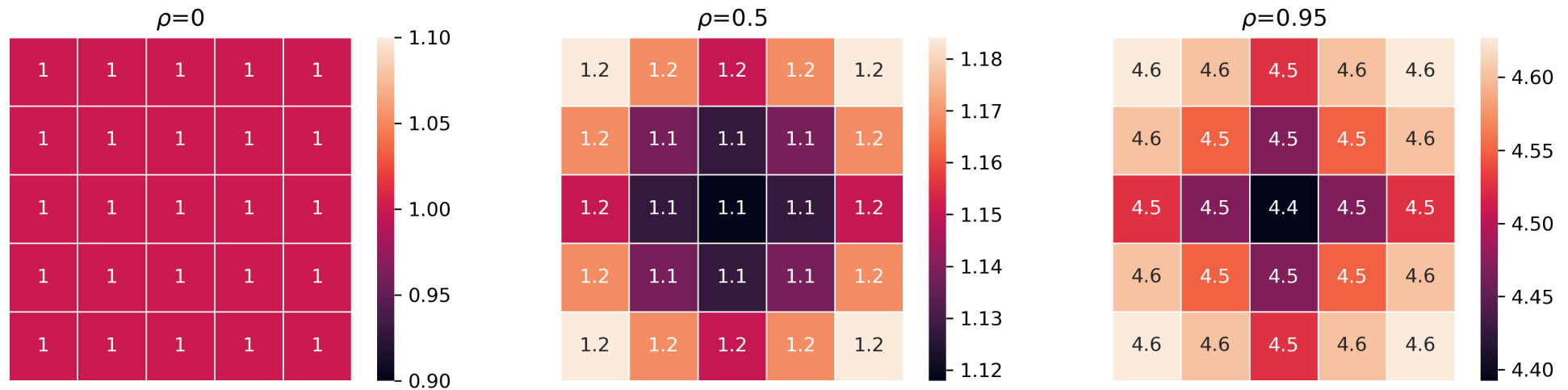
Why do we need this?



Simulating SAR:
Rook Contiguity, (5,5) grid

$$\mathbf{y} = (\mathbf{I} - \rho\mathbf{W})^{-1}\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

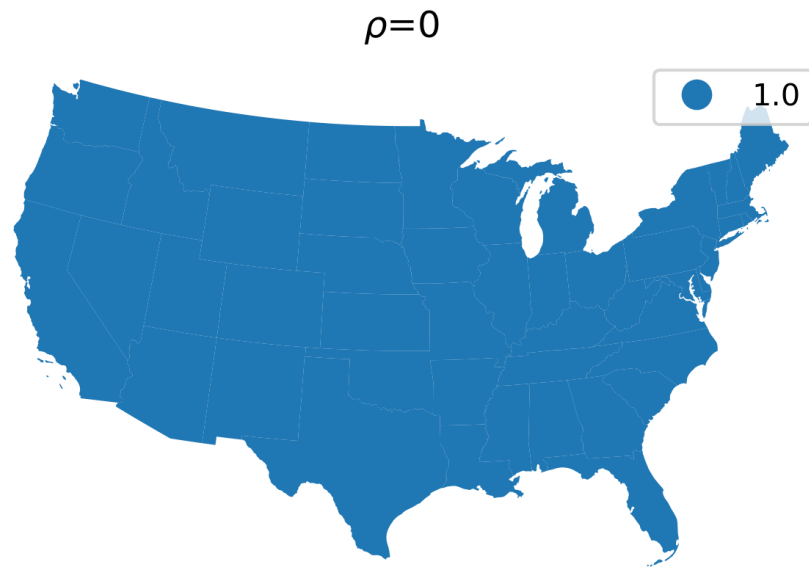
Why do we need this?



Marginal variance/std (1) Increases with ρ
(2) Varies across space

Why do we need this?

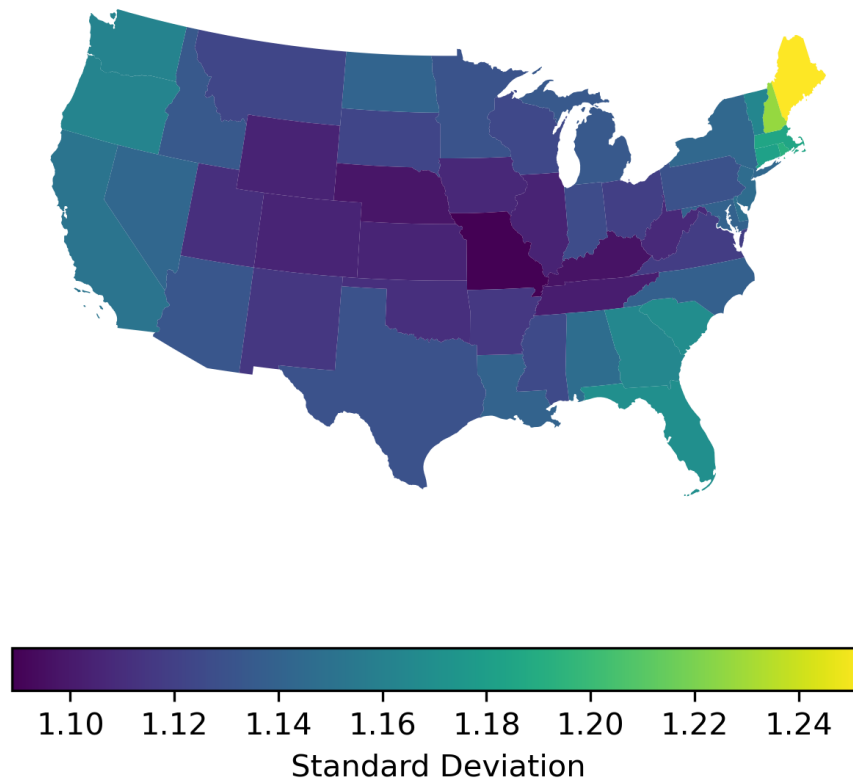
A real-world spatial structure



Why do we need this?

A real-world spatial structure

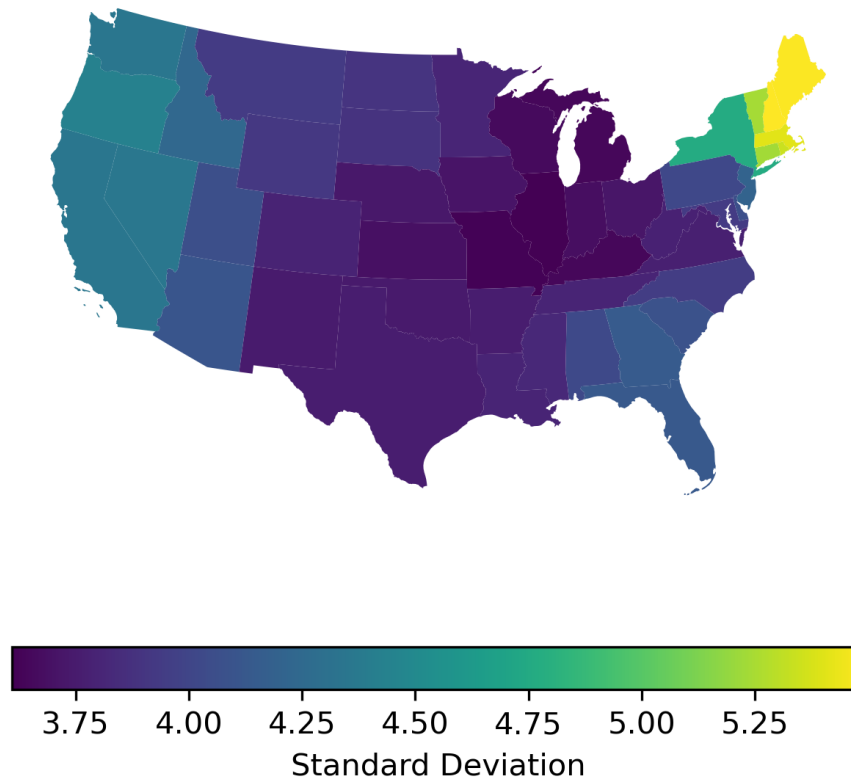
$\rho=0.5$



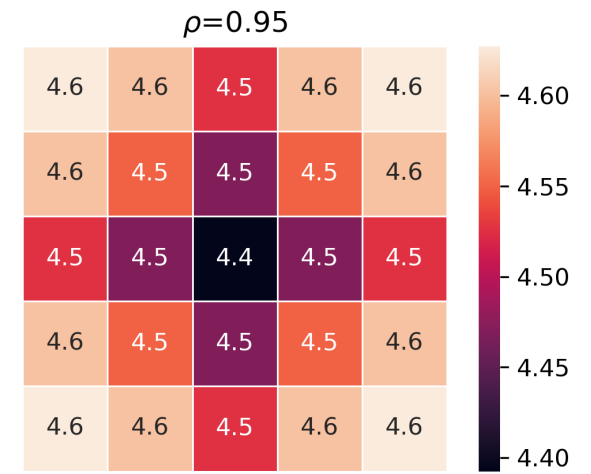
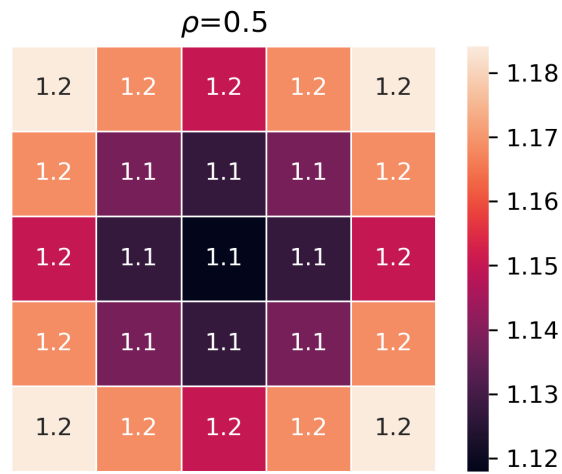
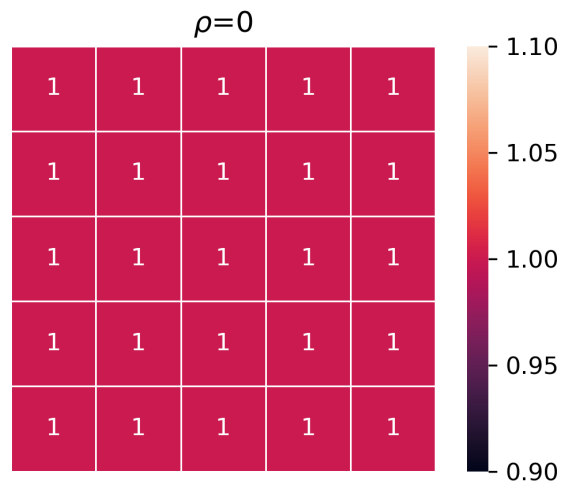
Why do we need this?

A real-world spatial structure

$$\rho=0.95$$

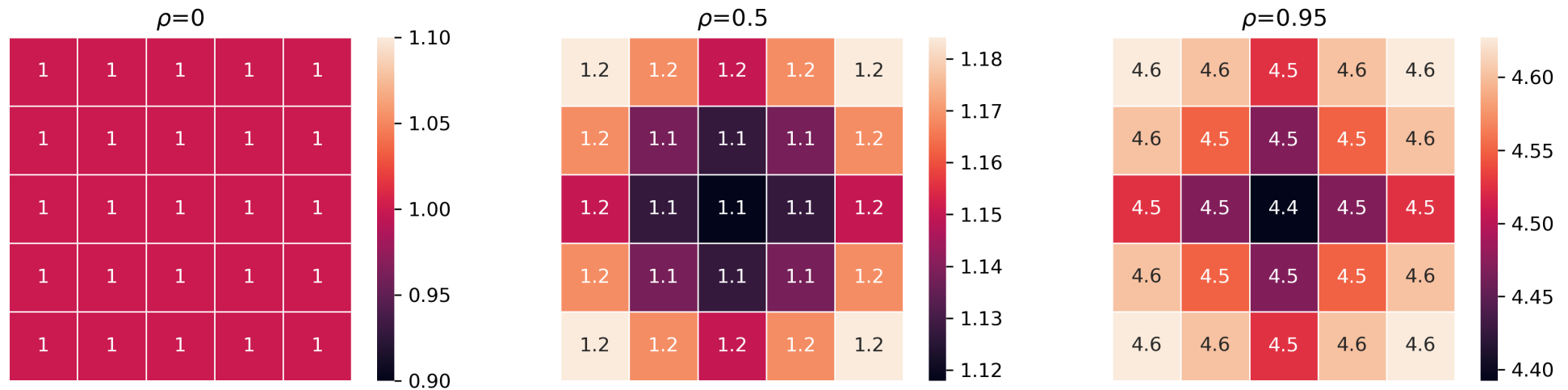


Why do we need this?



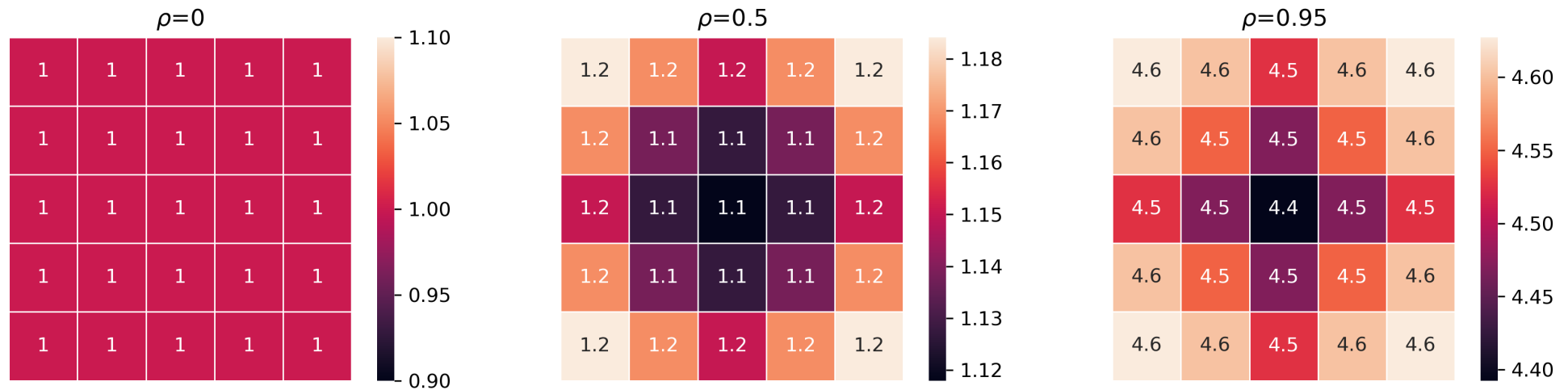
ρ is not a spatial *autocorrelation* parameter,
it's a spatial *covariance* parameter.

Why do we need this?



ρ is not a spatial *autocorrelation* parameter,
it's a spatial *covariance* parameter. Simulations over ρ
cannot separate **dependence** from **heteroskedasticity**.

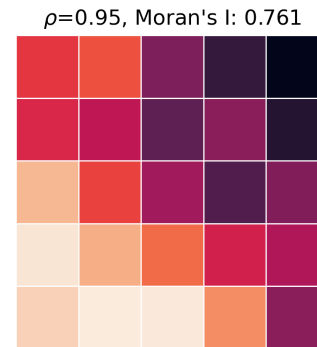
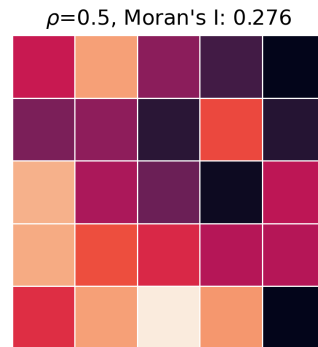
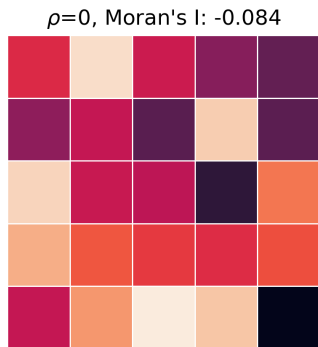
Why do we need this?



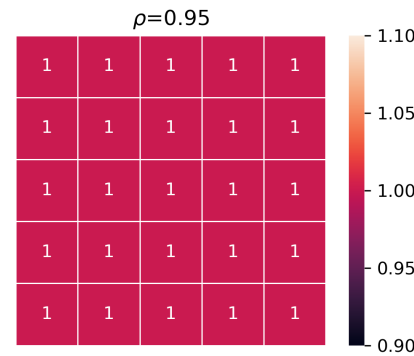
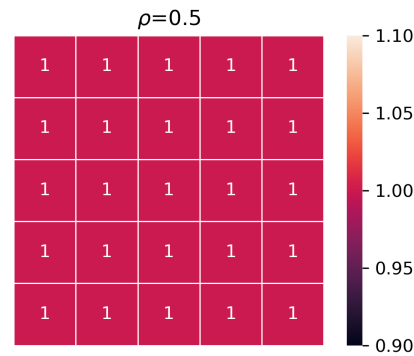
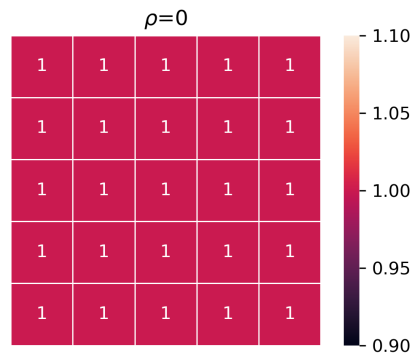
ρ is not a spatial *autocorrelation* parameter,
it's a spatial *covariance* parameter. Simulations over ρ
cannot separate **dependence** from **heteroskedasticity**.

Why do we need this?

Can we have one without the other?



SAR-style spatial autocorrelation



Constant/homoskedastic marginal variance

Gap and solution

Can we have one without the other?

SAR without heteroskedasticity

Three Monte Carlo experiments

Outline of the Talk

SAR without heteroskedasticity

From SAR to VSSAR

Three Monte Carlo experiments

Outline of the Talk

SAR:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}),$$

Reduced form:

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}).$$

Covariance matrix:

$$\boldsymbol{\Sigma}_y = \sigma^2 (\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{I} - \rho \mathbf{W}')^{-1}.$$

SAR for generating $\mathbf{y}$

Step 1: Decompose covariance into scale and correlation

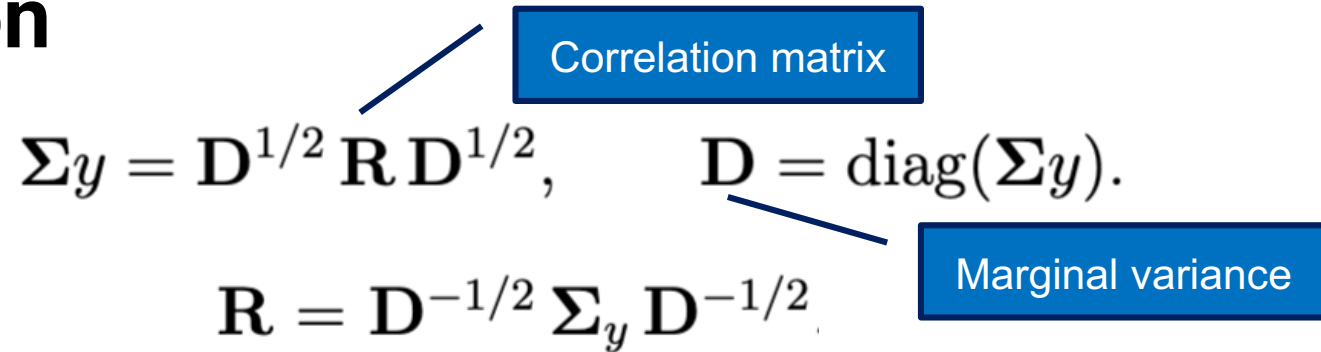
$$\Sigma y = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2}, \quad \mathbf{D} = \text{diag}(\Sigma y).$$

Correlation matrix

Marginal variance

Variance Stabilized SAR (VSSAR)

Step 1: Decompose covariance into scale and correlation

$$\Sigma_y = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2}, \quad \mathbf{D} = \text{diag}(\Sigma_y).$$


The diagram illustrates the decomposition of the covariance matrix Σ_y into a correlation matrix $\mathbf{R}$ and a diagonal matrix of marginal variances $\mathbf{D}$. The equation $\Sigma_y = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2}$ is shown, with a blue box labeled 'Correlation matrix' pointing to $\mathbf{R}$. Below it, the equation $\mathbf{D} = \text{diag}(\Sigma_y)$ is shown, with a blue box labeled 'Marginal variance' pointing to $\mathbf{D}$. The inverse relationship $\mathbf{R} = \mathbf{D}^{-1/2} \Sigma_y \mathbf{D}^{-1/2}$ is also provided.

$$\mathbf{R} = \mathbf{D}^{-1/2} \Sigma_y \mathbf{D}^{-1/2}$$

Variance Stabilized SAR (VSSAR)

Step 1: Decompose covariance into scale and correlation

$$\Sigma_y = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2}, \quad \mathbf{D} = \text{diag}(\Sigma_y).$$

Correlation matrix

$$\mathbf{R} = \mathbf{D}^{-1/2} \Sigma_y \mathbf{D}^{-1/2}.$$

Marginal variance

Step 2: Simulating with the correlation matrix $\mathbf{R}$

$$\mathbf{R} = \mathbf{L}\mathbf{L}^\top,$$

Cholesky factor

Variance Stabilized SAR (VSSAR)

Step 1: Decompose covariance into scale and correlation

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Correlation matrix

$$\mathbf{R} = \mathbf{D}^{-1/2} \Sigma_y \mathbf{D}^{-1/2}.$$

Marginal variance

Step 2: Simulating with the correlation matrix R

$$\mathbf{R} = \mathbf{L} \mathbf{L}^\top,$$

Cholesky factor

$$\gamma = \mathbf{L} \epsilon$$

iid standard normal distribution

Variance Stabilized SAR (VSSAR)

Step 1: Decompose covariance into scale and correlation

$$\Sigma_y = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2}, \quad \mathbf{D} = \text{diag}(\Sigma_y).$$

Correlation matrix

$$\mathbf{R} = \mathbf{D}^{-1/2} \Sigma_y \mathbf{D}^{-1/2}$$

Marginal variance

Step 2: Simulating with the marginal variance $\mathbf{D}$

$$\gamma = \mathbf{D} \varepsilon$$

iid standard normal distribution

SARSKE (SAR-Skedastic uncorrelated)

SAR without heteroskedasticity

From SAR to VSSAR

Three Monte Carlo experiments

- σ -convergence**
- OLS**
- Multiscale GWR (MGWR)**

Outline of the Talk

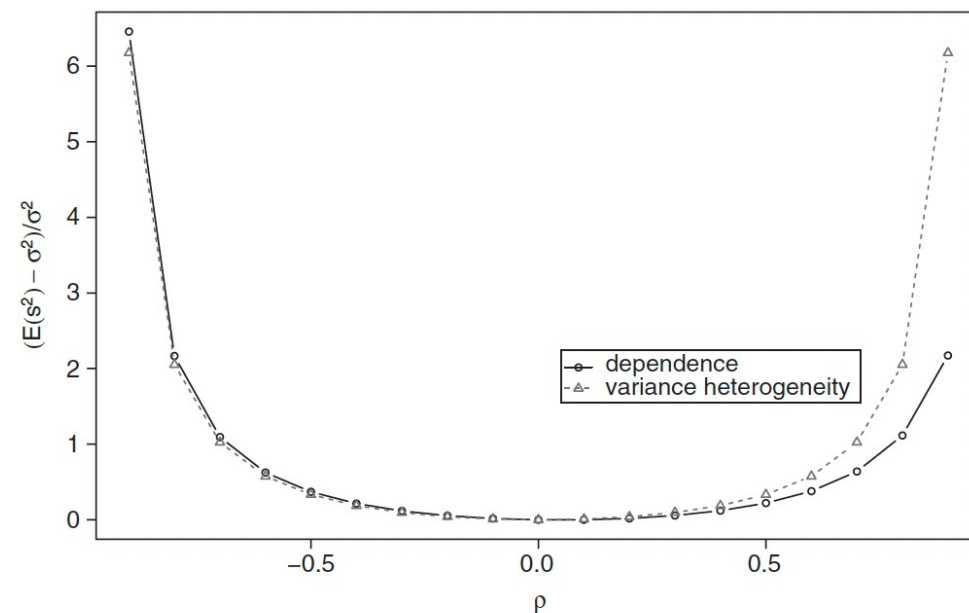
DGP	Spatial Autocorrelation	Spatial Heteroskedasticity
SAR	Yes	Yes
VSSAR	Yes	No
SARSKE	No	Yes

Three Data Generating Processes (DGPs)

Q: Are regions' incomes becoming more homogenous over time?

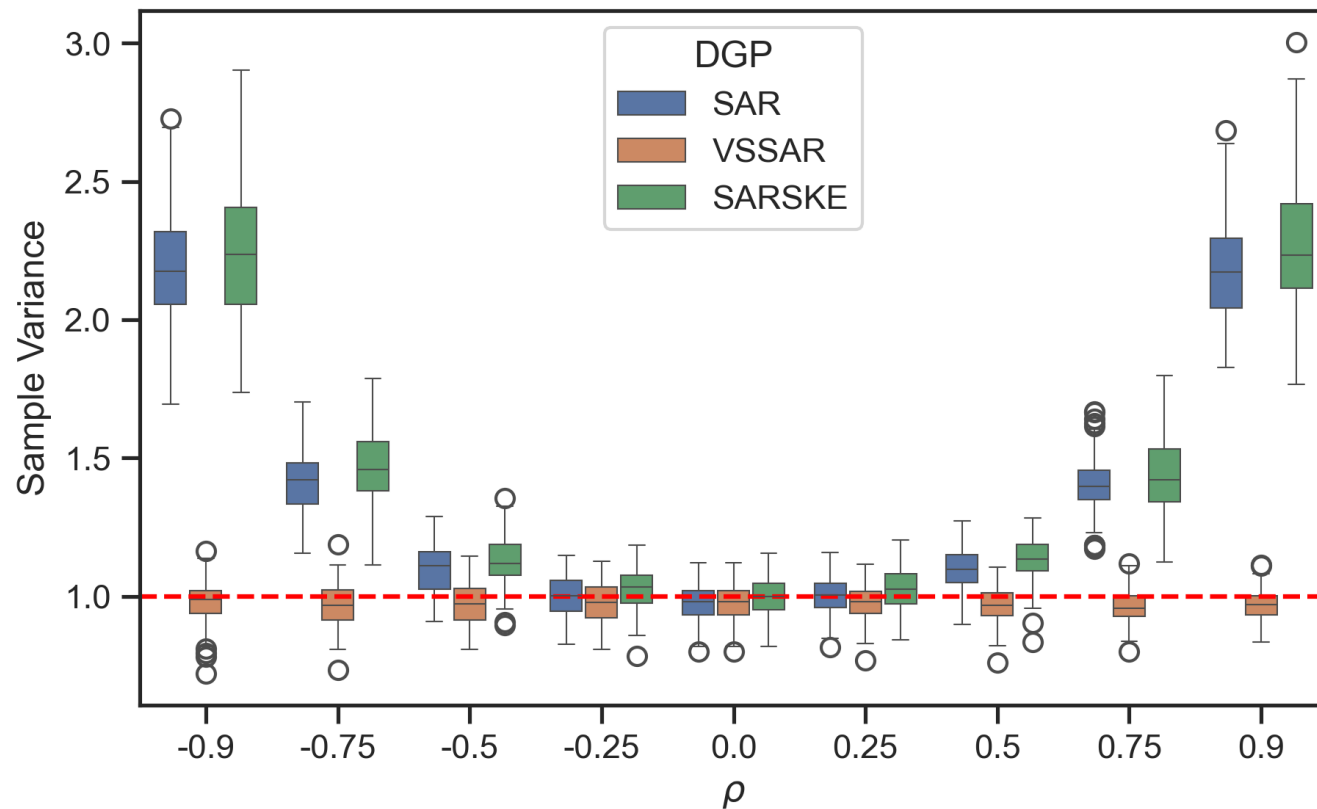
We know ρ leads to biased inference w.r.t sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2,$$



*Rey, S. J. and Dev, B. (2006). σ -convergence in the presence of spatial effects. *Papers in Regional Science*, 85(2):217–234.*

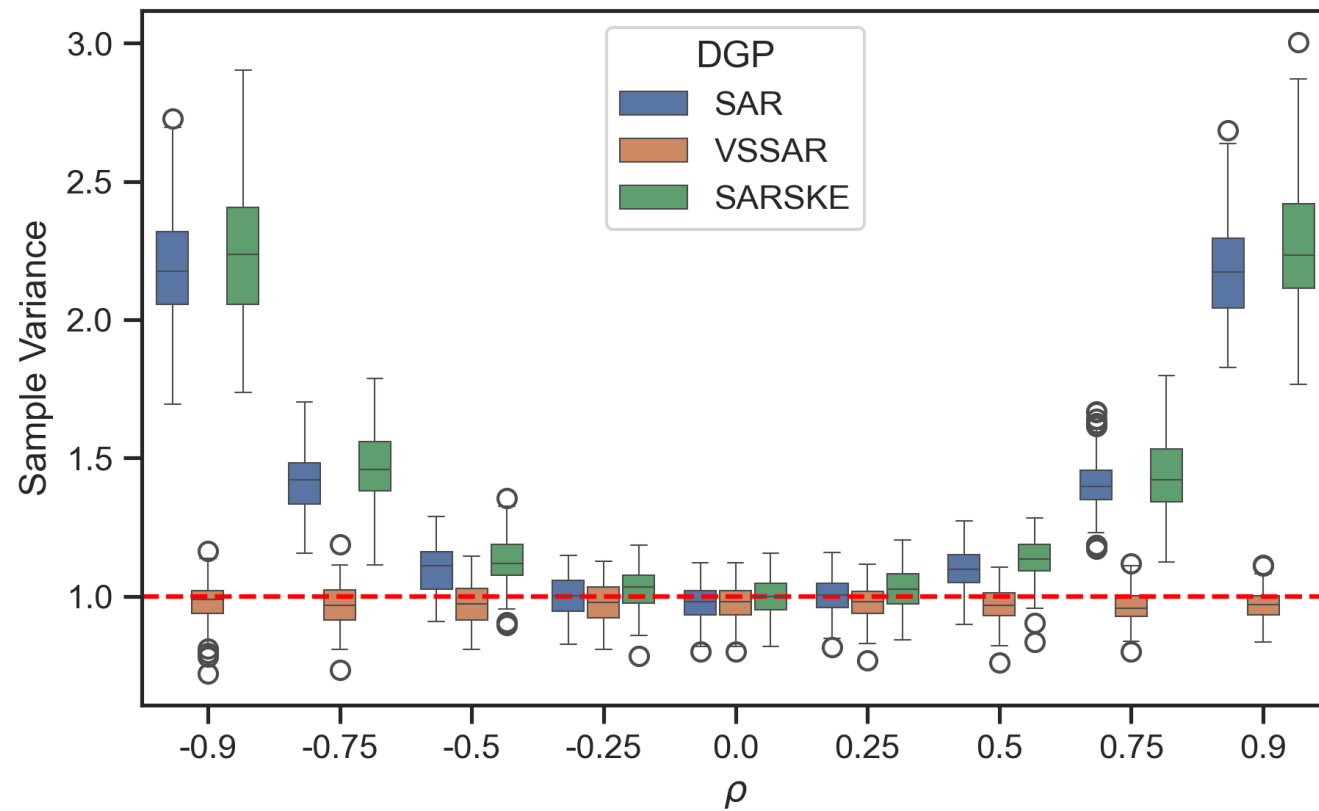
σ -convergence



Simulation design:

- (10, 10) regular grid
- Rook Contiguity
- $\rho \in [-0.9, 0.9]$
- 100 simulations per DGP

σ -convergence



Bias is caused by heteroskedasticity of SAR RVs, not SAR's spatial autocorrelation!

σ -convergence

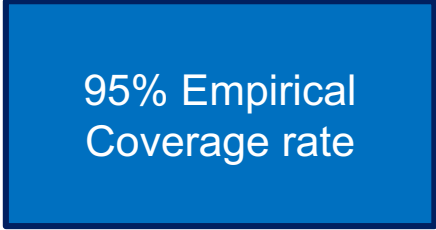
Q: Association between X and y in a simple linear model:

$$y = \alpha + X\beta + u$$

We know coverage rates decline as $|\rho|$ increases.

Simulation design:

- 1,000 spatial units, coordinates drawn from unit square.
- Delaunay spatial weights
- Single regressor $X_1 \sim \text{Unif}(0,1)$
- True parameters: $\alpha, \beta \sim N(0,1)$
- $\rho = 0.5, 0.95, 1,000$ replications



95% Empirical
Coverage rate

OLS estimator

Empirical coverage rates of 95% confidence intervals for the intercept (α) and slope (β))

	ρ	DGP	Coverage (α)	Coverage (β))
1	0	i.i.d. baseline	0.956	0.944
2	0.5	SAR	0.098	0.922
3	0.5	VSSAR	0.121	0.926
4	0.5	SARSKE	0.895	0.958
5	0.95	SAR	0.008	0.618
6	0.95	VSSAR	0.037	0.336
7	0.95	SARSKE	0.908	0.952

OLS estimator

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OLS estimator

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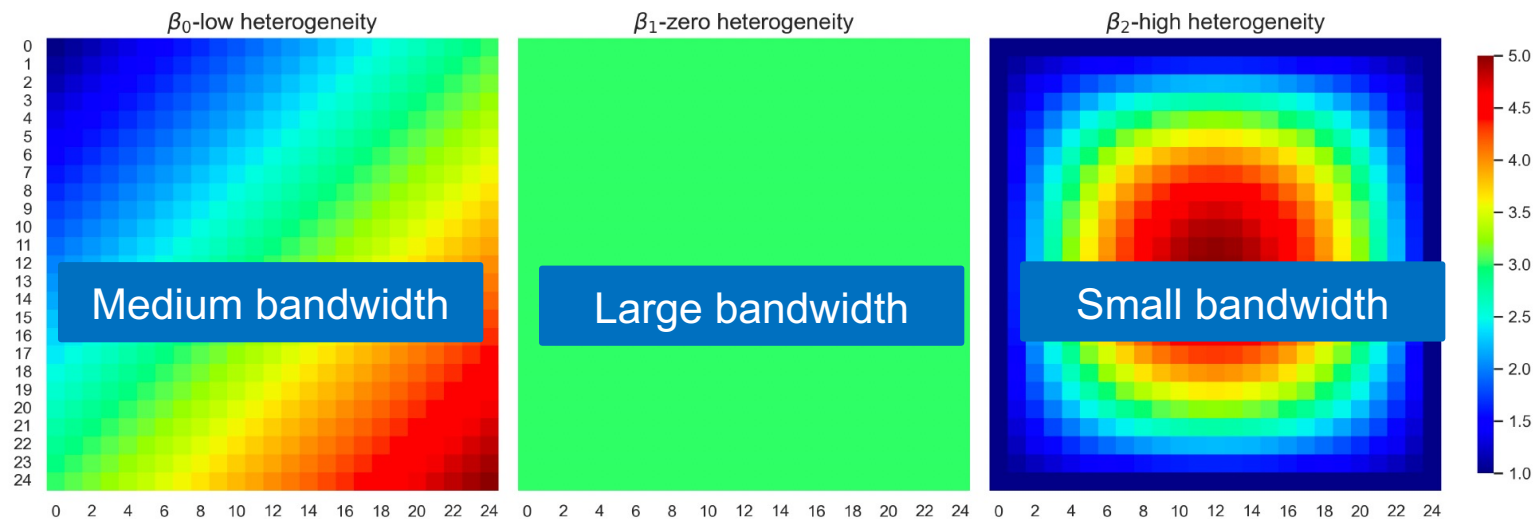
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Coverage reduction overwhelmingly caused by spatial autocorrelation, not heteroskedasticity

OLS estimator

MGWR: GAM for process spatial heterogeneity

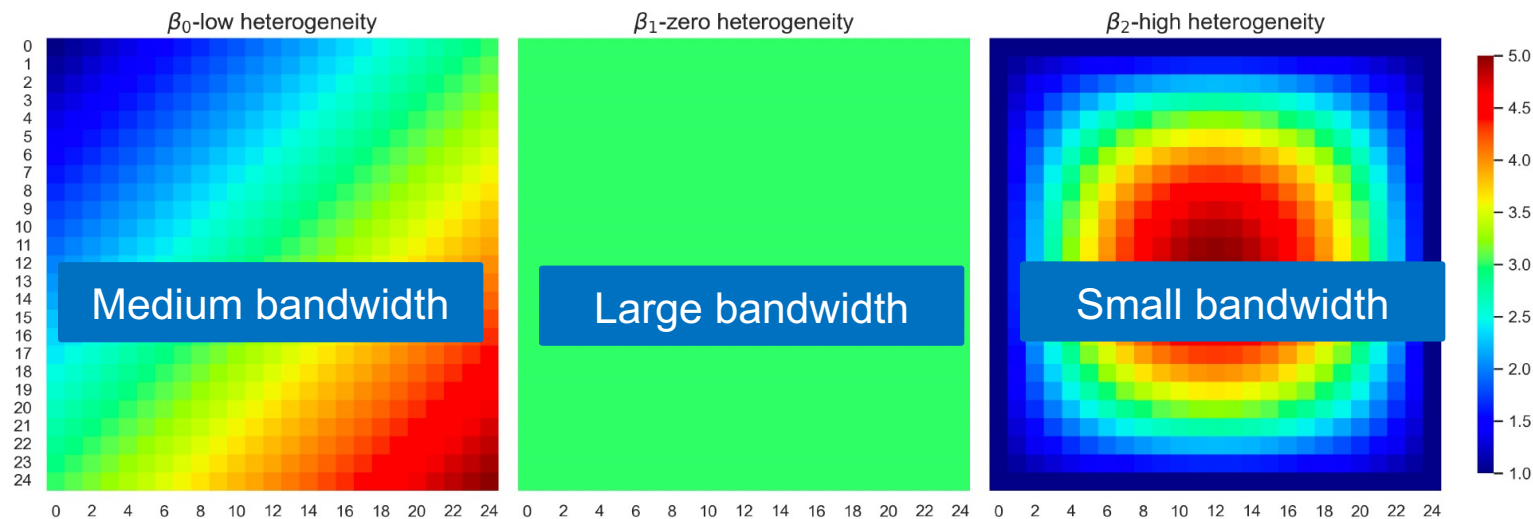
$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^K \beta_k(u_i, v_i)x_{ik} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$



MGWR

MGWR: GAM for process spatial heterogeneity

$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^K \beta_k(u_i, v_i)x_{ik} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$



MGWR

Bandwidth is typically a function of the local variability of the surface, and ρ^+ is associated with larger bandwidths

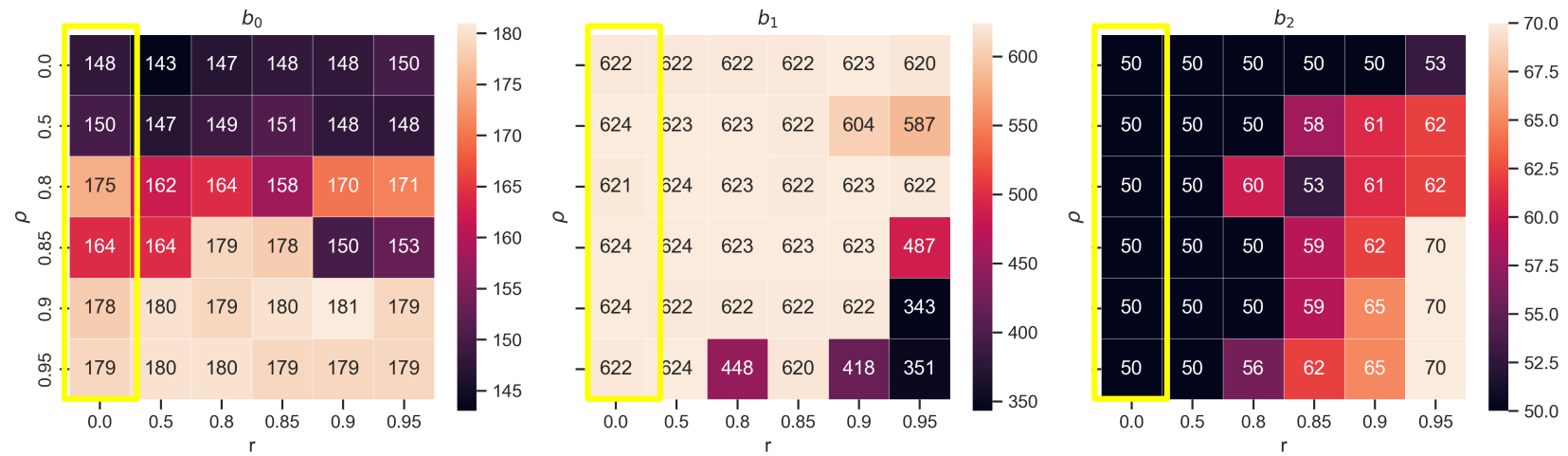
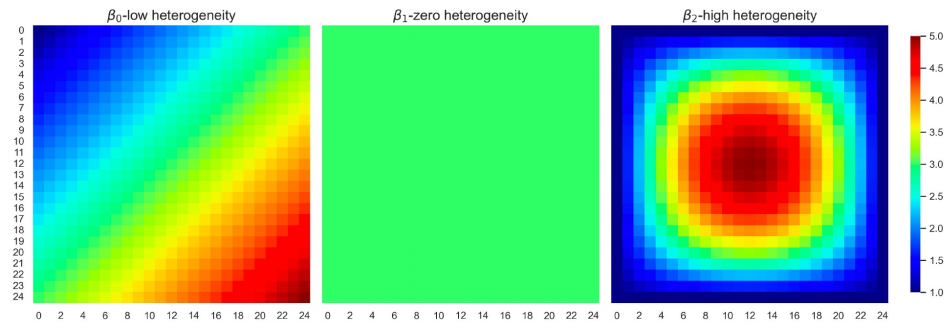
Kang & Oshan (2025): studied how two types of correlation impacted MGWR

- Global correlation between two Covariates (multicollinearity issue)
- Spatial autocorrelation in Covariates – commonly encountered in empirical studies

SAR process for generating spatially autocorrelated covariates X
Applied normalization to ensure fixed global mean/variance

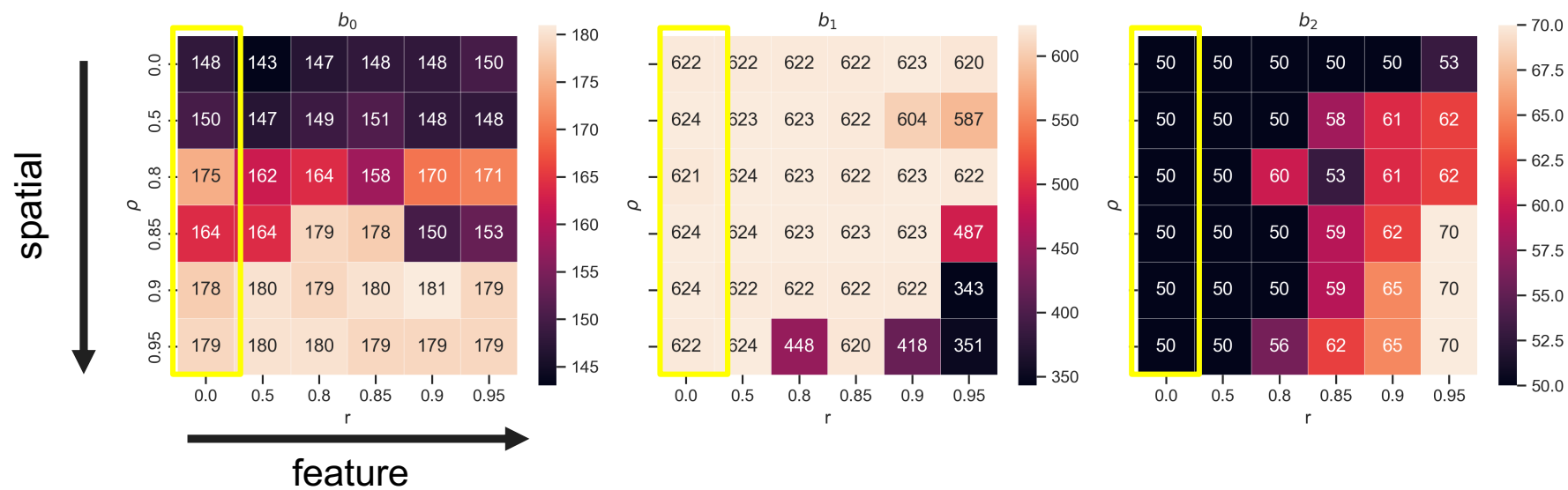
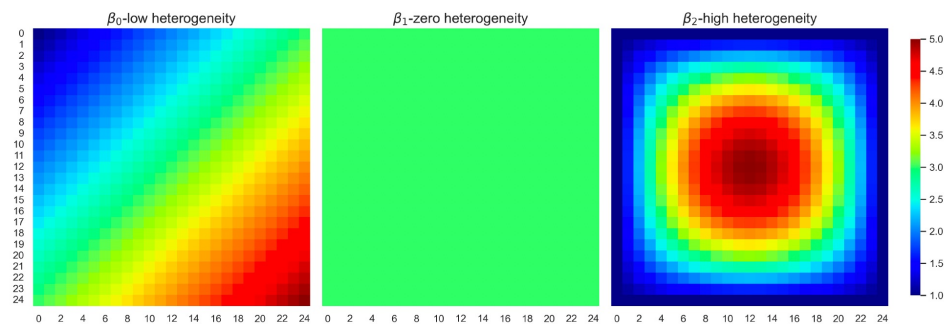
MGWR

Kang, W. and Oshan, T. M. (2025). Scale and correlation in multiscale geographically weighted regression (MGWR). *Journal of Geographical Systems*.



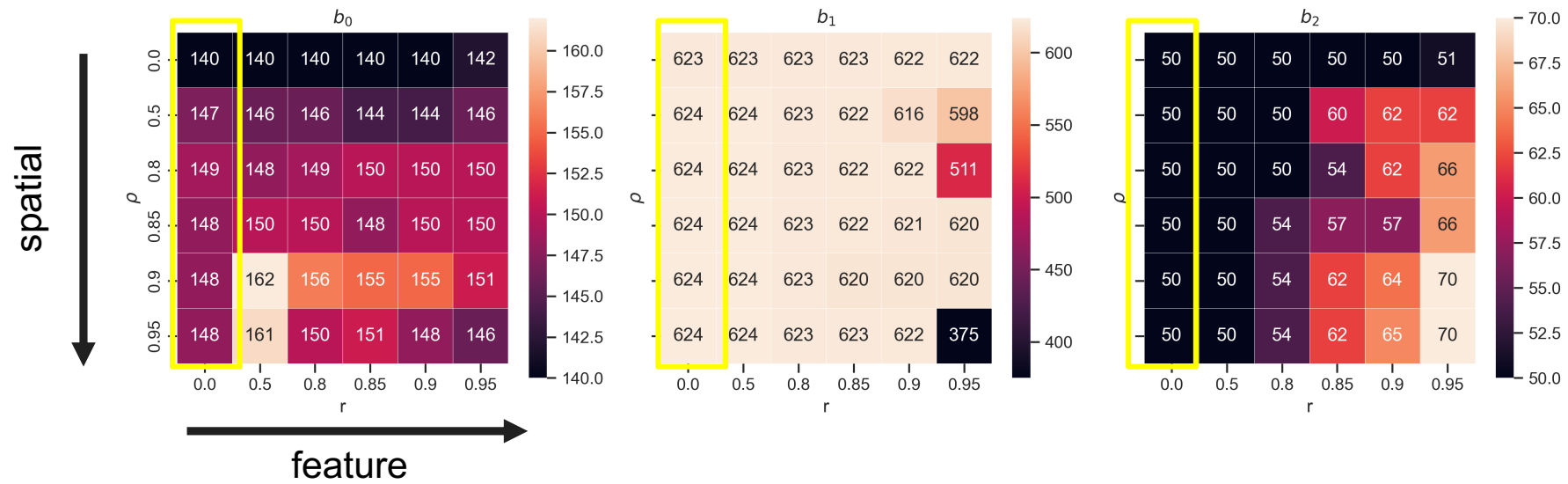
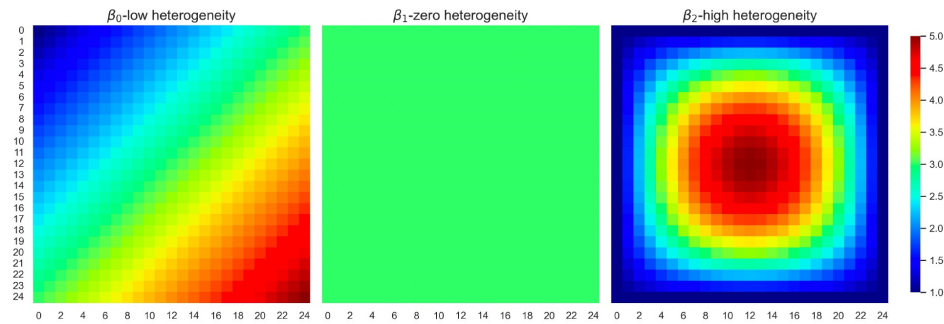
MGWR-median bandwidth

Standard SAR w/ post-normalization



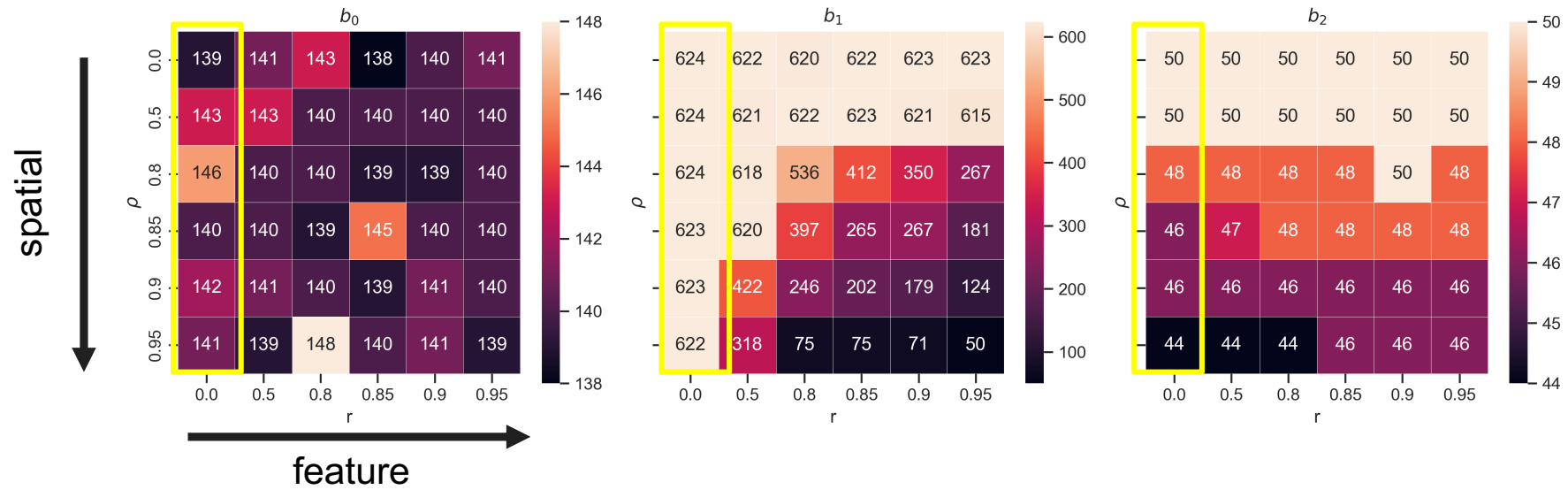
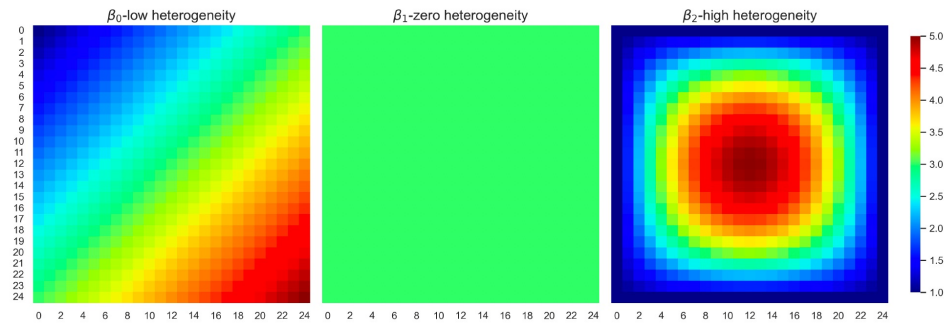
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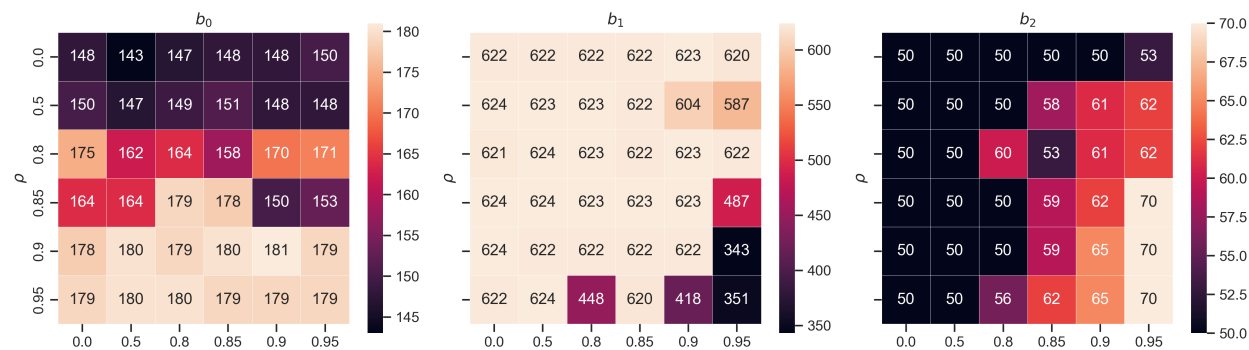
VSSAR (only autocorrelation)



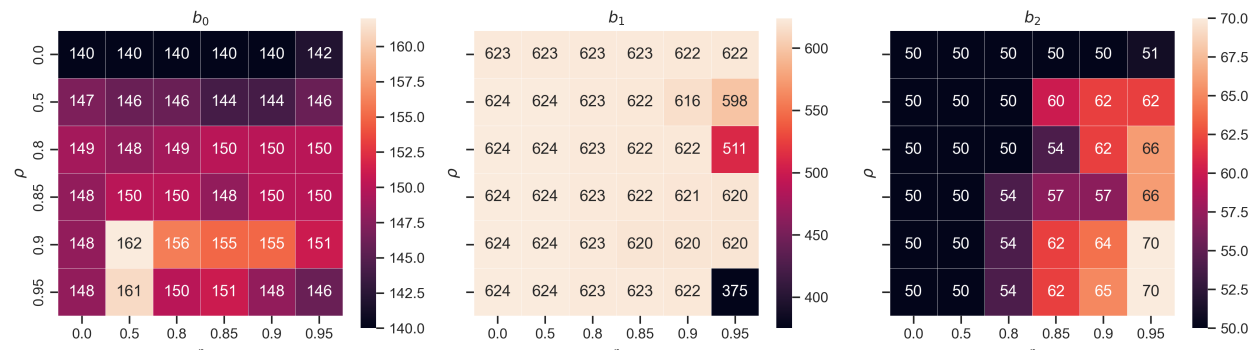
MGWR-median bandwidth

SARSKE (only spatial heteroskedasticity)

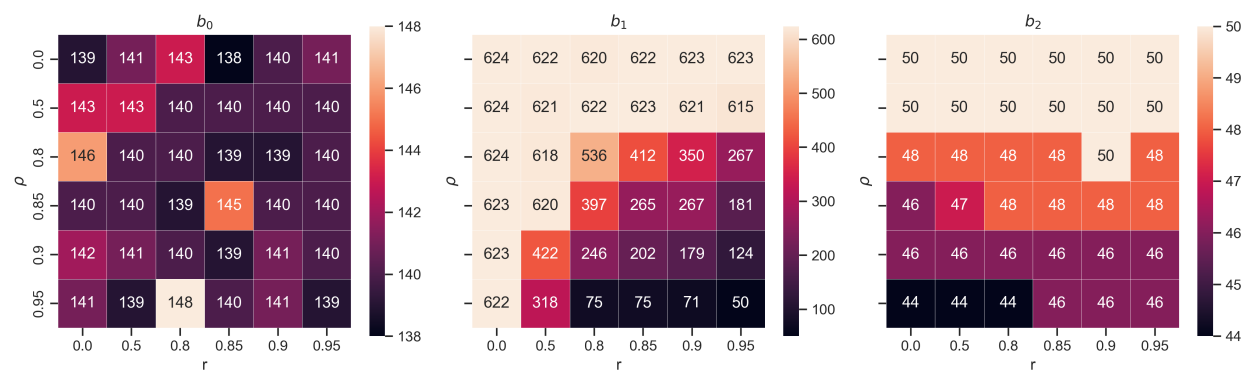
SAR and post-normalization



VSSAR
(autocorrelation)

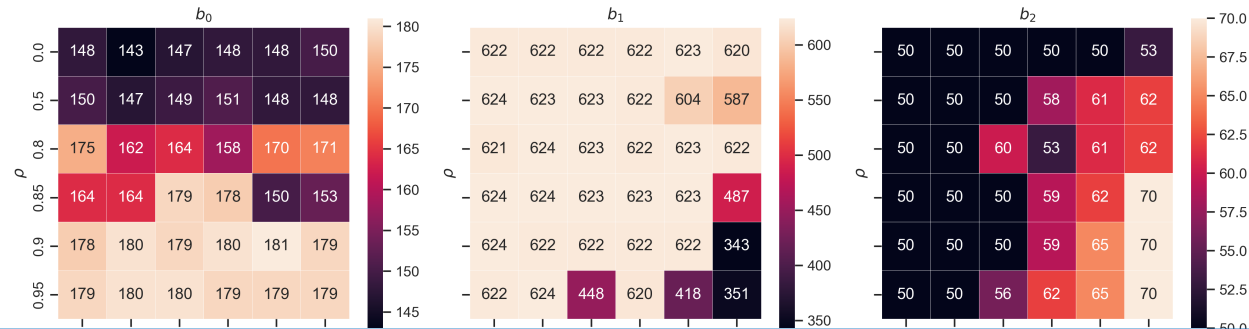


SARSKE
(heteroskedasticity)

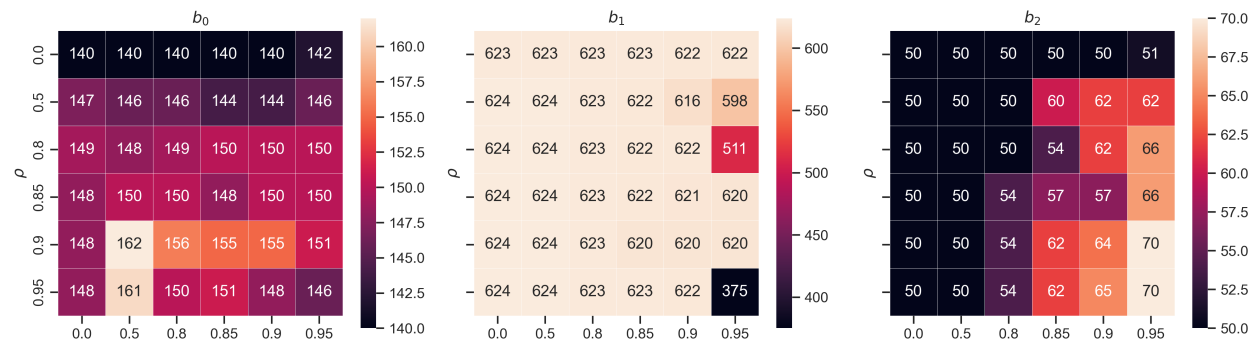


MGWR-median bandwidth

SAR and post-normalization

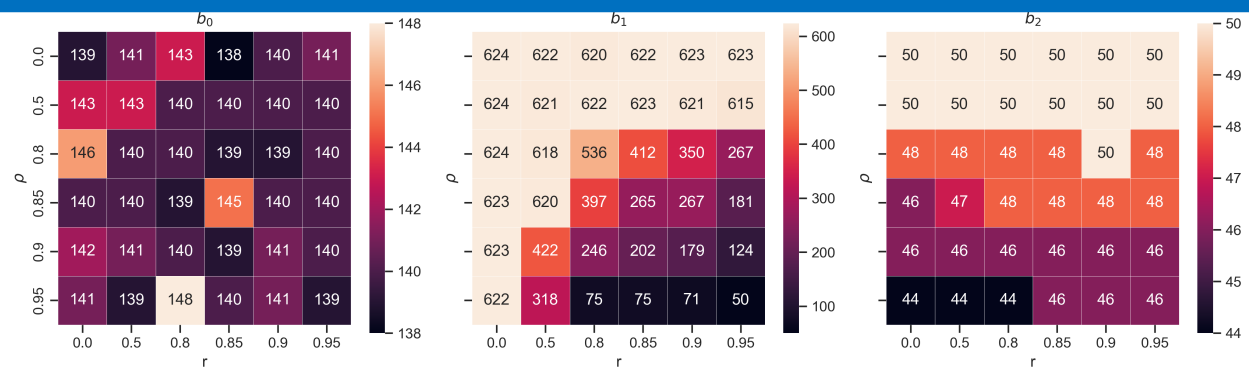


VSSAR (autocorrelation)



**Most
stable
over ρ**

SARSKE (heteroskedasticity)



MGWR-median bandwidth

SAR Process's ρ is a spatial covariance parameter

→ affects both correlation & variance.

∴ SAR simulation cannot separate dependence from heteroskedasticity

VSSAR cleanly isolates correlation from heteroskedasticity, allowing us to clarify:

- heteroskedasticity alone matters for σ -convergence
- dependence dominates for OLS estimator size
- dependence matters most for local spatial models (MGWR)

Discussion

Future work should use stabilized SAR to separate spatial dependence from heteroskedasticity!

Thank you!
Questions?

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